# Lab Report: Text classification with neural networks

## 1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. The dataset was split into 80 percent training data and the rest for testing.

## 2 Results

Figure 1 shows the test accuracy of GRU by dataset.

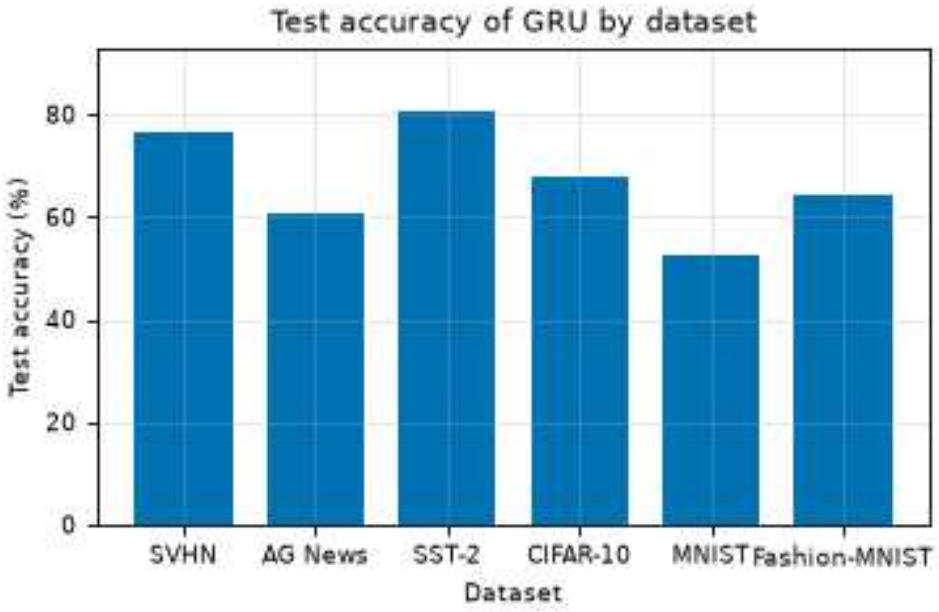



Figure 1: Test accuracy of GRU by dataset for the models in this experiment.

As shown in Figure 1, GRU scores 22.6 points higher on SST-2 than on SVHN. GRU achieves 66.3% test accuracy on Fashion-MNIST (Figure 1). As shown in Figure 1, GRU reaches 60.8% test accuracy on AG News. GRU achieves 86.4% test accuracy on CIFAR-10.

Figure 2 shows the accuracy during training.

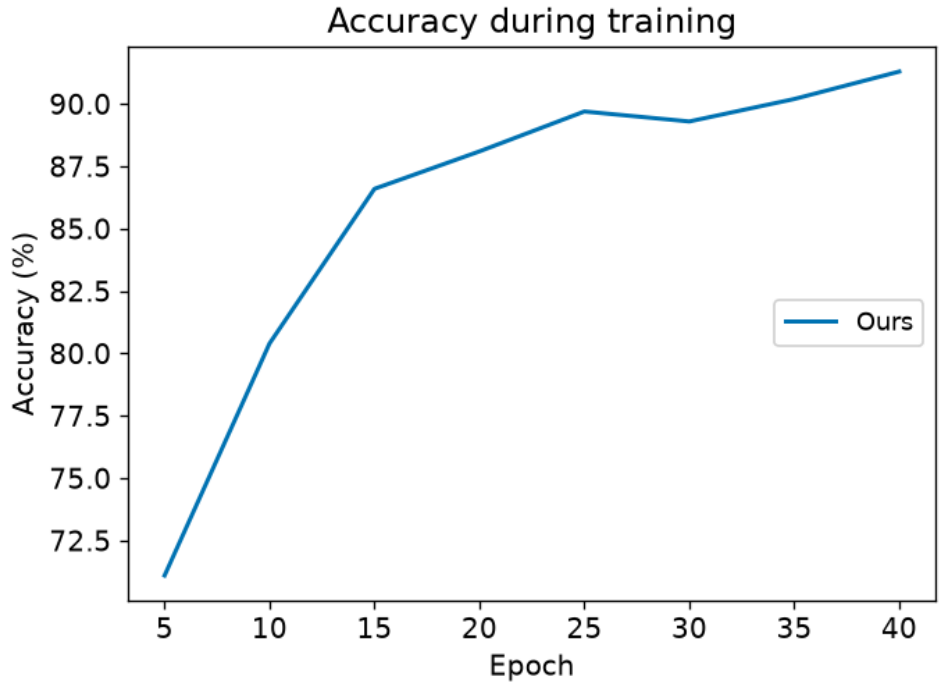



Figure 2: Accuracy during training for the models in this experiment.

From epoch 20 to epoch 40, the accuracy of Ours improves by a relative 4.1%. Ours attains a accuracy of 80.4% at epoch 10. At epoch 35, Ours achieves about 90% accuracy. Ours scores 1.6 points higher on 40 than on 25.

## 3 Discussion

XGBoost reaches 81.8% accuracy on SVHN. The differences between the models are discussed in the next section.